Universität Zürich[UZH]

Department of Chemistry

DSA 105 Lab Automation and Chemical Data Science in Practice

Information

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DSA 105 Projects

— Projects employ all contents of the DSA minor

— Can be started either in autumn or spring semester

— 2 semesters time to finish the project

— No fixed dates – project work done at home (data part) and upon arrangement in the lab (experiments)

— Done in small groups of 2 or 3 people

How to book the module:

1. Apply to johannes.schoergenhumer@chem.uzh.ch (or in person)
2. State which semester you plan to start
3. And if already known: other group members and topic of interest (can also be done later, once you have seen the data)
4. You will get access to the DSA105 repository
5. If you haven't done so, state your team members and the project you chose

DSA 105 Key idea and steps

Work from home
Lab work upon arrangement

1. You get access to research datasets from UZH – choose one of the projects
2. Process and clean the data, analyse thoroughly, visualise (Content DSA103)
3. Think about possible research questions
4. Project meeting with project supervisor to focus on some ideas to follow up with ML models
5. Use ML models for further data analysis or predictions, etc. (Content DSA104)
6. Project meeting with project supervisor to discuss digital output to validate, test, expand in the lab
7. Lab work to experimentally test hypothesis, collect more data, … (arrange with project supervisors) (Content DSA102)
8. Review of experimental outcome and discuss with project supervisors
9. Hand in deliverables

DSA 105 Deliverables and grading

Deliverables:

— Well commented python script or Jupyter notebook for the data part (can be referred to in the report)

— Short written report about the story from the dataset to the experimental results (main focus on the discussion of the latter)

Grading:

— Experimental work

— Deliverables

What is expected from these team projects:

— Collaborative effort (fair workload distribution among the team members)

— High degree of self-reliant project planning and execution

— Adhere to good practices on the data part (documentation (markdown, comments, etc), high-quality graphs / plots, use Git/GitHub, …)

— At least one ML model needs to be employed. Choice of models needs to be justified, models should be optimised.

— Deliverables: Scripts/notebooks have to run without errors in a well defined virtual environment (e.g. DSA104 or your own one)

Currently available project topics / datasets

1) Biotransformation dataset (Eawag, supervisor: Jasmin Hafner):
   - Dataset on the rate of biotransformation of micropollutants in activated sludge
   - About 800 samples
   - Unpublished research data

2) Multimodal spectroscopy dataset (HTEL, supervisor: Johannes Schörgenhumer):
   - Dataset comprising simulated NMR, IR and MS spectra
   - Rather large (~790k samples)
   - Published dataset (https://zenodo.org/records/14770232)

3) Suzuki coupling dataset (HTEL, supervisor: Johannes Schörgenhumer):
   - small, classic reaction dataset (~300 entries)
   - extracted from an ELN, exclusively manual experiments
   - Unpublished research data